

FINVISION: A MICROFINANCE LOAN MANAGEMENT SYSTEM

Course: IST 659 M003

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INTRODUCTION

- FinVision aims to improve operational efficiency by streamlining loan processes and reducing processing times.
- It enhances data visibility through real-time insights into loan performance and client interactions via intuitive dashboards, supporting data-driven decision-making.
- FinVision also facilitates client engagement by understanding borrower demographics and preferences while strengthening risk management through effective monitoring of loan performance.

OVERVIEW

- Comprehensive Loan Management
- Client Management
- Repayment Monitoring
- Financial Insights
- Risk Management
- Data Visualization
- Operational Efficiency

PROJECT FEATURES

- **Loan Portfolio Tracking:** This will track the number of active loans, the loan amount disbursed and loan statuses.
- **Client Demographics and Segmentation:** This will organize and visualize the client data based on demographic factors such as age, gender location, income and loan history. It also takes into consideration factors such as loan size, business type and credit history.
- **Repayment and Delinquency Tracking:** This feature monitors loan repayment schedules including dates and amounts. It tracks repayments, missed payments and overdue loans offering insights into repayment and delinquency rates.
- **Financial Performance Monitoring:** This system tracks key financial metrics such as interest income, fees, operational costs and overall profitability. It provides dashboards that compare revenue from loans against operational expenses, giving an overall picture of financial health

LOAN PORTFOLIO OVERVIEW DASHBOARD

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- **Total Loan Amount Disbursed:** This metric shows the cumulative total of all loans disbursed over a specific period, providing insights into the overall lending activity.
- **Loan Types Distribution:** This metric categorizes the loans by type, such as agricultural, small business, or personal loans, helping us understand which products are most popular or profitable.
- **Loan Status Distribution:** This visual representation breaks down the loans by their current statuses, such as Active, Overdue, Closed, or Defaulted.
- **Active Loans Count and Value:** This metric indicates the number of currently active loans and their total value, reflecting the institution's current exposure and engagement with borrowers.
- **Top Borrowers List:** This lists the borrowers who have received the highest loan amounts, allowing us to recognize key clients and assess potential risks or opportunities for cross-selling.

REPAYMENT AND DELINQUENCY DASHBOARD

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- **Repayment Rate:** The repayment rate measures the percentage of loans that are being repaid on time versus those that are overdue. This metric provides insight into the overall health of the loan portfolio.
- **Repayment Trends Over Time:** This metric tracks changes in repayment patterns over time, enabling the identification of seasonal trends or the impact of policy changes on repayment behavior.
- **Delinquency Rate:** The delinquency rate shows the percentage of loans that are past due, indicating how many loans are not being repaid as scheduled. This helps identify areas of concern in loan performance.

CLIENT DEMOGRAPHICS DASHBOARD

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- **Borrower Demographics (Age, Gender, Income Level):** Understanding the demographic profile helps tailor services to meet client needs.
- **Client Segmentation (Loan Size, Credit History):** allows targeted marketing and risk assessment. Segmenting clients helps in personalizing loan offerings. (?)
- **Client Growth Over Time:** This metric tracks the growth of the client base over time, distinguishing between new and returning clients. It provides insight into the effectiveness of client acquisition strategies.

FINANCIAL PERFORMANCE DASHBOARD

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- **Interest Income (AUM):** This metric tracks the total income generated from interest on loans disbursed, providing insight into the profitability of the lending portfolio.
- **Charges (Non AUM):** These metrics evaluate the cumulative Non AUM we will receive like processing fees , penalties , setup fee .

PROPOSED SCHEMA

Dashboards:

1. Loan Portfolio Overview Dashboard

1. Total Loan Amount Disbursed
2. Active Loans Count and Value
3. Loan Status Distribution
4. Loan Types Distribution
5. Top Borrowers List

2. Financial Performance Dashboard

1. Interest Income (AUM)
2. Charges (Non AUM)

DATABASE STRUCTURES

Tables:

1. Users
2. Loans
3. Repayments
4. Loan Officers
5. Transactions

Review - Under Discussion

- Tables under review
 - Collateral
 - Penalty
 - Credit history
- Visualization Tool : Superset , Power BI ?

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YOU**

LOAN OFFICER PERFORMANCE DASHBOARD

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- **Loan Disbursement by Officer:** This metric tracks the total amount of loans disbursed by each loan officer over a specific period.
- **Loan Recovery by Officer:** This metric measures the total amount of loan repayments collected by each officer, reflecting their effectiveness in managing collections.
- **Delinquency Rates by Officer:** This metric tracks the percentage of loans that are overdue or in default, categorized by the loan officer responsible for those accounts.
- **Client Interactions:** This metric tracks the percentage of loans that are overdue or in default, categorized by the loan officer responsible for those accounts.